Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering three-dimensional fluorescence from fixed-focus images requires measurements that distinguish structures at different depths. We learn a phase plate for angle-scanned excitation by optimizing this discrimination independently of reconstruction. A plane wave illuminates a fixed, defocused plate from varying angles, producing depth-dependent excitation while the specimen and detection focus remain stationary. Training minimizes weighted squared correlations between detector-integrated depth signatures, without specimen examples or a reconstruction solver. We evaluate the same plate with total variation (TV) and plug-and-play (PnP) reconstruction using pretrained and neuron-trained denoisers. In simulations of mouse pyramidal neuron volumes at eightfold measurement compression and matched nominal excitation dose, axial-profile Wasserstein error decreases by 28–50 % and centroid error by 30–54 %. The objective also yields 13–20 % lower profile error than the tested coherence and Fisher-information surrogates under matched training budgets. In controlled filament crossings, the qualifying separation falls from 3.84 *to* $2.56\,\mu\mathrm{m}$ *with TV and pretrained PnP, while remaining unchanged with neuron-trained PnP. The results demonstrate improved axial recovery across the tested priors and identify where lower average error does, and does not, translate into structural separation.*

1. Introduction

Three-dimensional fluorescence microscopy must distinguish structures that overlap in projection but occupy different depths. When their measurement responses are similar, a sharp reconstruction may still place fluorescence at the wrong axial locations. Reconstruction priors help select a volume, but their success depends on the depth information present in the images [2, 7]. We investigate whether learning the excitation illumination can improve axial recovery across reconstruction methods.

Our system encodes depth through angle-scanned excitation. A plane wave illuminates a fixed, defocused phase plate, producing a spatially structured field in the specimen. Changing the incident angle shifts the excitation differently at different depths, so structures at different axial positions experience distinct sequences of illumination. A camera records one fluorescence image per angle at a fixed detection focus. Only the input angle changes; the specimen, plate and detection focus remain stationary.

We learn the phase plate by reducing similarity between the measurement responses of voxels at different depths and the same lateral position. Each response includes the entire imaging chain: structured excitation, fluorescence detection and camera-pixel integration. The objective uses these physical responses directly, requiring neither specimen examples nor a reconstruction network. It builds on coherence-based sensing design and decoder-independent optical optimization [6, 18, 25], specializing them to excitation-side depth encoding. Unlike training through a chosen decoder [11, 15], this formulation leaves the reconstruction prior outside optical training.

We evaluate the learned plate on simulated measurements of mouse pyramidal neuron volumes using total variation (TV) and plug-and-play (PnP) reconstruction with pretrained and neuron-trained denoisers. We measure both the recovered axial intensity profiles and their centroids, together with image quality. Controlled filament crossings test whether lower depth error also supports separation of overlapping structures. These evaluations distinguish improvements shared across reconstruction methods from tasks that still depend on the prior. We further compare optical objectives under the same training budget and examine reduced measurement counts, noise and alignment errors.

Our contributions are:

1. **Excitation-side axial encoding.** We develop a fixed-focus imaging model in which an angle scan through a defocused phase plate encodes depth, with vectorial propagation and voxel-integrated detection responses.
2. **Learning from optical responses.** We optimize the plate using a weighted squared-correlation objective on depth signatures, without specimen examples or a reconstruction solver.
3. **Evaluation of axial recovery across priors.** We test the same learned optics with TV and two PnP decoders, evaluating full depth profiles, centroid accuracy and con-

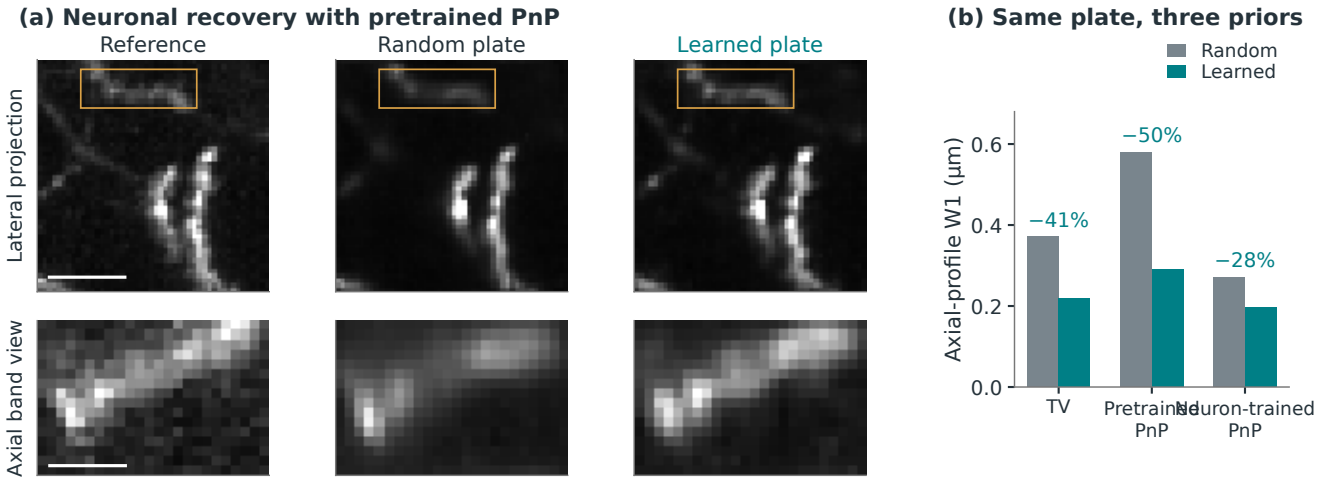


Figure 1. **Learning the illumination improves axial recovery across priors.** (a) Full lateral projections and matched axial band views with pretrained PnP, the prior with the largest mean gain ($C = 8$, draw 0). The crop and outlined band were selected from reference structure, before viewing reconstruction gains. Band views maximize over the outlined 5.1 μm strip; depth increases downward. Each row shares one linear display scale. (b) Mean axial-profile W1 over the 60 test crops, using the same learned plate with all three priors. Centroid errors and paired intervals are in Tab. 1. Bars: 10 μm (projections), 5 μm (band views, true aspect).

trolled structural separation under matched acquisition conditions.

2. Related work

Depth encoding in fluorescence microscopy. Computational fluorescence microscopes encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D recover volumes from depth-dependent detection responses [1, 4, 25]. On the excitation side, Yeh *et al*. [26] combine translated speckles with axial scanning. Pascucci *et al*. [16] exploit axial speckle decorrelation for volumetric recovery from lateral scans, with fluorescence saturation improving resolution. Angular scanning through a scattering layer has also enabled two-dimensional fluorescence recovery [27], while tilted pattern projection produces depth-dependent lateral shifts [9]. These precedents establish structured excitation as a source of depth information. We learn a fixed phase plate for angular illumination scanning, keeping both the specimen and detection focus stationary.

Learning optical encoders. End-to-end methods optimize optics jointly with reconstruction networks [8, 15, 22]; Kellman *et al*. [11] optimize illumination through a specified reconstruction algorithm. Decoder-independent design instead uses properties of the measurements. Shechtman *et al*. [21] optimize Fisher information for emitter localization, and Miniscope3D [25] reduces cross-coherence between detection PSFs across depth. Gram-matrix and coherence objectives also have a long history in sensing-matrix design [5, 6]. IDEAL and IDEAL-IO [14, 18] optimize estimated mutual information using fitted measurement-distribution models. Our method belongs to this measurement-driven family. Its objective compares detector-integrated voxel responses across illumination angles, targeting different depths at each lateral location. It requires no specimen examples, fitted specimen distribution or reconstruction solver. We compare particular coherence and Fisher-information surrogates under the same optical training budget; this does not rank the broader design frameworks.

Volumetric reconstruction and priors. TV regularization promotes piecewise-smooth structure [20], while PnP methods use denoisers as implicit priors [10, 23]. Multi-slice fusion combines denoisers along complementary orientations of a shared volume [12]. We adapt this framework to fluorescence reconstruction and evaluate both pretrained and neuron-trained priors. Evaluating the same optics with TV and PnP tests whether their benefit depends on a particular learned reconstruction prior, while recognizing that a plausible volume alone does not establish unique recovery [2, 7].

3. Optical model and learning

Acquisition. A tilted plane wave illuminates a fixed phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, where $\mathbf{r} = (x, y)$. For direction t, its incident field is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^\top \mathbf{r}}$, with transverse direction cosine $\mathbf{s}_t$ and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ in the simulated medium of refractive index one. We sample 256 distinct forward directions with $0.5 \le \|\mathbf{s}_t\| \le \sqrt{3}/2$; the angular grid is specified in the supplement. After phase modula-

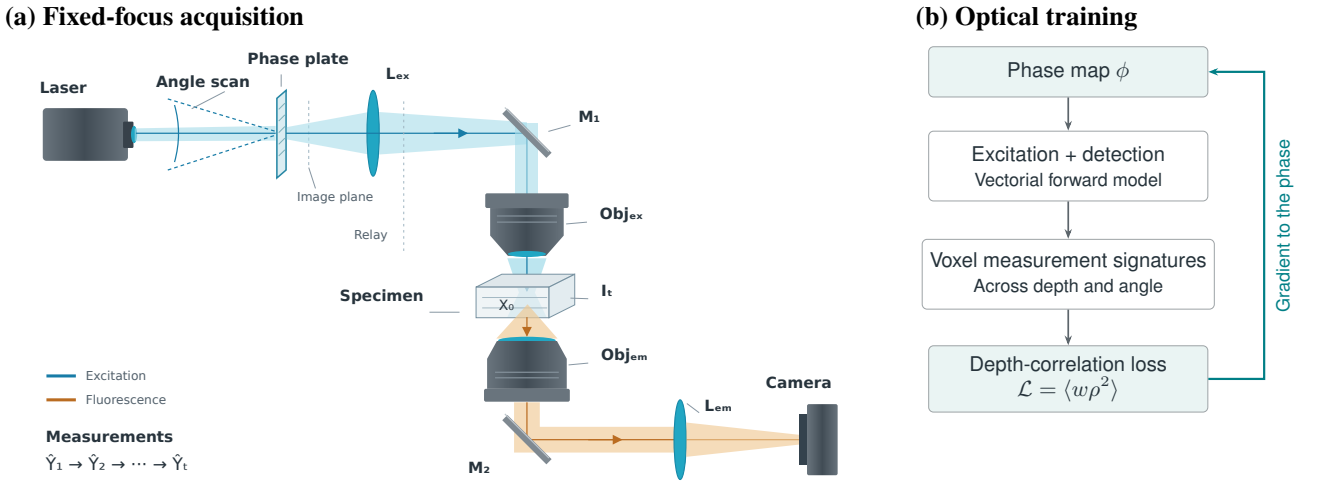


Figure 2. **Learning the illumination while keeping the microscope fixed.** (a) An angle-scanned plane wave passes through a defocused phase plate; the camera records fluorescence at a fixed focus. Blue and orange indicate excitation and emission. (b) The plate is optimized through the optical model using depth-signature correlations. Reconstruction is evaluated separately.

tion, angular-spectrum propagation carries the field over distance D to an intermediate illumination image plane. The objective pupil then filters this field before a vectorial Richards–Wolf relay [13, 19] produces the specimen excitation, $I_t(\mathbf{r}, z) = \sum_{c\in\{x,y,z\}} |E_{t,c}(\mathbf{r}, z)|^2$. The vectorial model accounts for the high excitation NA; the incident polarization is along x. All coordinates are specimen-equivalent. The fixed-focus camera records one image per direction, so acquisition is sequential.

Why the angle encodes depth. Propagation after the phase plate converts its phase structure into spatially varying intensity. A change in illumination angle shifts this pattern differently at different depths. In the ideal paraxial thin-screen model, changing the transverse direction cosine by $\Delta\mathbf{s}$ gives a lateral displacement $\Delta\mathbf{r} \approx d\,\Delta\mathbf{s}$ after propagation over distance d. Each depth therefore experiences a different sequence of excitation intensities during the scan. Defocusing the plate places the shear pivot outside the specimen. This relation explains the encoding principle; the vectorial relay and finite pupil determine the actual patterns, including departures from pure translation.

Image formation. Let $X_0(\mathbf{r}, z)$ be the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$ [26]. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z')X_0(\mathbf{r}', z')I_t(\mathbf{r}', z')\,\mathrm{d}^2\mathbf{r}'\,\mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity PSF and Ω is the specimen volume. Pixel p, occupying region D_p, records $\bar{y}_{t,p} = |D_p|^{-1}\int_{D_p} g_t(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$. Fluorescence adds incoherently before pixel averaging. Excitation multiplies the specimen before detection blur, allowing fine illumination structure to affect measurements even when the camera cannot resolve that pattern directly.

Voxel-integrated operator. The optical field can vary within a reconstruction voxel. We therefore simulate the optics on a finer grid and integrate its responses over each reconstruction voxel. If B is the fine-grid measurement operator and U replicates each reconstruction coefficient over its fine-grid cells, the reconstruction operator is $\mathsf{A} = \mathsf{BU}$. This is an exact restriction of the discretized model to a piecewise-constant basis.

Sub-voxel specimen structure still produces approximation error. For a fine-grid specimen $\mathbf{x}_f$ and its voxel-averaged representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$,

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{UR}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Our simulated measurements are generated before this voxel averaging, retaining $\boldsymbol{\delta}_{\mathrm{disc}}$ rather than setting it to zero.

Linear fluorescence permits efficient operator assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells and $g_{p,i}$ is cell i's detection response, including its integration weight. The responses $g_{p,i}$ are computed once and reused across angles.

After this precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF \log F)$ for constructing columns through separate FFT-based forward simulations: T illumination patterns, P camera samples, F fine-grid cells and N reconstruction voxels.

Learning the plate. At lateral position $\mathbf{r}$ and depth z, the corresponding column of A describes how a unit voxel contributes to all camera measurements. Let $\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z}$ be this signature restricted to a sampled angle set $\mathcal{T}$, centered over its angle–pixel entries and normalized to unit norm. We minimize

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}} \left[\frac{\sum_{z \neq z'} w_{zz'} \left((\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z})^{\top} \mathbf{a}^{\mathcal{T}}_{\mathbf{r},z'} \right)^2}{\sum_{z \neq z'} w_{zz'}} \right]. \quad (4)$$

The fixed weights give selected reflected-depth pairs twice the weight of other pairs; their indices are specified in the supplement. Squaring penalizes correlation of either sign. Signature normalization makes the loss invariant to uniform optical-power scaling, while deposited excitation dose is matched separately at evaluation.

Gradients pass through excitation propagation, the vectorial relay and detector integration to the plate phase. Training requires no specimen examples or reconstruction solver. The loss is a surrogate for depth discrimination: reducing pairwise similarity does not ensure unique or stable recovery of a volume. We test its effect using the independent reconstruction methods in Sec. 4.

4. Reconstruction with fixed priors

We evaluate the same learned plate with three reconstruction methods: TV, pretrained PnP and neuron-trained PnP. None participates in optical training. Each method uses the operator for its encoder, with the prior held fixed across encoders. Scalar reconstruction parameters are selected separately for each encoder and measurement budget on held-out validation volumes.

Total variation. The non-learned baseline minimizes $\frac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \|\nabla \mathbf{x}\|_{2,1}$ subject to $\mathbf{x} \in [0,1]^N$, where $\|\nabla \mathbf{x}\|_{2,1}$ is isotropic three-dimensional TV [20]. We solve this problem using ADMM [3].

Plug-and-play reconstruction. PnP combines the measurement model with denoisers acting as implicit priors [10, 23]. Following multi-slice fusion [12], we denoise complementary slices of a shared volume. For orientation $o \in \{xy, yz, zx\}$, P_o rearranges the volume into slices processed independently by a two-dimensional denoiser, giving $\mathcal{D}_o = \mathsf{P}_o^{-1} \mathcal{D} \mathsf{P}_o$.

PnP-ADMM alternates a quadratic measurement-consistency update with denoising updates. Let $\mathcal{J}$ index

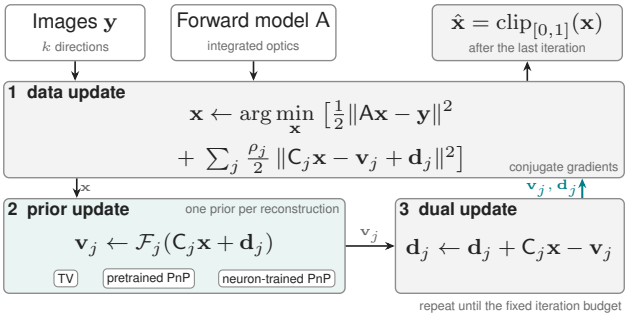


Figure 3. **Reconstruction with interchangeable priors.** A quadratic data update couples the images and forward model to a shared volume. Prior and dual updates feed back through ADMM. Each run uses one prior family; the networks remain fixed across encoders. Branch transforms C_j are identities for PnP and the gradient/identity for TV. TV enforces its box constraint in the loop; PnP clips the final estimate.

the active prior branches, with penalties ρ_j, auxiliary volumes $\mathbf{v}_j$ and scaled dual variables $\mathbf{d}_j$. The volume update is

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \sum_{j \in \mathcal{J}} \frac{\rho_j}{2} \|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2, \quad (5)$$

solved by conjugate gradients. Each branch then applies its denoising operator $\mathcal{F}_j$:

$$\begin{aligned} \mathbf{v}_j^{k+1} &= \mathcal{F}_j(\mathbf{x}^{k+1} + \mathbf{d}_j^k), \\ \mathbf{d}_j^{k+1} &= \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}. \end{aligned} \quad (6)$$

These are PnP updates; a generic denoiser need not be the proximal map of an explicit regularizer.

Pretrained PnP uses a natural-image DnCNN [28] along the three slice orientations. *Neuron-trained PnP* replaces the slice denoiser with a neuron-trained 2D DnCNN and adds a neuron-trained 3D denoising branch. Thus, the latter changes both the training distribution and volumetric prior. An optional ℓ_1 branch is weighted by validation and is inactive in every main-comparison setting. The networks are trained on separate neuronal data and remain fixed during reconstruction. Both PnP variants start from normalized back-projection and clip the final estimate to $[0,1]$; TV enforces its box constraint during optimization. Network architectures, branch weights, stopping rules and parameter selection are given in the supplement.

5. Experimental setup

Simulation settings. We simulate linear fluorescence in a homogeneous, non-scattering medium. Excitation and emission use a common wavelength of 532 nm as a modeling approximation, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has 0.32 μm features and lies

$6.6\,\mu\mathrm{m}$ before the intermediate illumination image plane in specimen-equivalent coordinates. Fields are evaluated at $0.08\,\mu\mathrm{m}$ lateral and $0.32\,\mu\mathrm{m}$ axial spacing, then binned to a $0.32\,\mu\mathrm{m}$ analysis grid. The reconstruction uses $0.64\,\mu\mathrm{m}$ isotropic voxels. Measurements are generated on the analysis grid before reconstruction-voxel averaging, as in Eq. (2).

Volumes and measurements. Each reconstruction contains $16 \times 48 \times 48$ voxels ($N = 36{,}864$), spanning $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image has 12×12 samples. Acquiring $k = 256$, 64 or 32 illumination directions gives measurement compression $C = N/(144k) = 1$, 4 or 8. For $C > 1$, we use five randomly drawn direction subsets, paired across encoders. Compression refers to the number of scalar measurements relative to reconstruction coefficients; it is not a measured acquisition-speed gain.

Data and validation. Reference volumes come from public two-photon stacks of mouse pyramidal neurons [24], min–max normalized and resampled to the analysis grid. The test set comprises six cells from three animals. Two crops per cell were selected by a reference-only structure score before reconstruction; eight further crops per cell were sampled at random after reconstruction parameters were fixed. The resulting 60 crops support the main cross-decoder evaluation. Other experiments use the 12 prespecified crops unless stated otherwise. Neuron-trained denoisers use 459 crops from a fourth animal. Scalar reconstruction parameters are chosen separately for each encoder and measurement setting on two vascular crops from another animal. No test volume is used for training or tuning.

Optical controls and dose. We compare the learned plate with its random initialization, two further random phase plates, defocused binary amplitude masks, an overfilled plate-free angular scan and an in-focus DMD with random binary patterns. The relay, pupil, detector and operator assembly are shared. Excitation is simulated beyond the specimen field to suppress beam-edge encoding. Each encoder is normalized to the same mean excitation over the specimen and all 256 directions. With k directions, the photon model is $n_i \sim \mathrm{Poisson}(\kappa_0\, 256\bar{y}_i/k)$. Exposure κ_0 is set separately for each specimen using the random initialization to give 1000 expected photons per camera pixel per image at the mean signal for $k = 256$, then shared across encoders. This matches deposited dose for the full scan and in expectation over random subsets; we call it *matched nominal dose*. Two optical training seeds vary the sampled positions and angles from the same initial phase. The supplement specifies the aperture, noise variants and tuning grids.

Axial and image-quality metrics. Our primary depth metric is the axial Wasserstein-1 error, which compares complete intensity profiles [17]. At each reference-signal column c, we normalize the reference and nonnegative reconstructed profiles to unit mass, with cumulative distributions F_c and $\hat{F}_c$. Then

$$E_{\mathrm{W1}} = \sum_c \pi_c\, \Delta z \sum_{i=0}^{Z-2} |F_c(i) - \hat{F}_c(i)|, \qquad (7)$$

where $\Delta z = 0.64\,\mu\mathrm{m}$ and π_c is proportional to reference column brightness, summing to one. We include columns brighter than 10 % of the reference maximum. Centroid error is the corresponding weighted absolute difference between axial intensity centroids. W1 detects profile differences that a centroid can conceal, but does not measure intensity or lateral fidelity; we therefore also report PSNR and foreground SSIM.

Paired comparisons. Encoders are paired on specimens, direction subsets, nominal dose and noise protocol. We first average draws within each crop, then report mean errors and relative changes in these means. Confidence intervals resample the six cells, keeping their crops together (10,000 bootstrap replicates). These intervals account for within-cell dependence but remain approximate for six cells from three animals.

Controlled structure tests. We also simulate crossing fluorescent filaments over a range of axial gaps, using three lateral crossing geometries, two axial centers and five measurement draws. Successful recovery requires matched axial peaks in columns carrying at least half of the reference crossing signal, together with correct depth ordering of the arms. We report the smallest tested gap at which success reaches 80 % at both centers and persists for larger gaps. A complementary measurement-only test compares neuronal volumes with alternatives preserving their lateral projections and column centroids while merging selected axial profiles. Full definitions are in the supplement.

6. Results

6.1. Depth discrimination before reconstruction

We first verify the physical depth code without learning. Angular scanning through the defocused phase plate produces depth-dependent lateral shifts, with a shear pivot near its conjugate plane. Cross-correlation estimates follow the geometrical trend, with departures from ideal translation caused by the finite pupil. In point-emitter tests with unknown brightness and 32 illumination directions, the random plates misassign depth in 1.5–1.7 % of cases, versus about 5 % for amplitude masks and 25 % for the tested in-focus DMD.

The amplitude masks remain competitive with random phase plates in neuronal centroid error; phase modulation is not intrinsically superior to amplitude modulation. The supplementary optical controls report all realizations and defocus settings.

Learning changes how the excitation is distributed across angles (Fig. 4). At the preset field-center position, each depth receives more than 10 % of its peak response in 33 % of the directions on average, compared with 98 % for the random plate. The mean cosine similarity of these temporal responses falls from 0.83 to 0.41. Across the field, the mean absolute correlation of the full detector-integrated depth signatures falls from 0.84 to 0.48. The temporal example illustrates depth-selective modulation; the field-wide statistic evaluates the complete signatures used in the objective.

We also test whether this change distinguishes a specific axial rearrangement of neuronal fluorescence. For each eligible crop, we merge selected axial profiles while preserving their column brightness and centroids. After fitting a global brightness scale, the learned plate increases Poisson measurement divergence from these alternatives by a factor of 10.0 (95 % CI: 8.7–11.8), with an increase in all 59 eligible crops. The result concerns these alternatives; it does not establish uniqueness among all volumes. Neither this test nor the optical loss uses a reconstruction prior.

6.2. Axial recovery across reconstruction priors

At eightfold measurement compression, the learned plate lowers mean axial-profile error by 41 % with TV, 50 % with pretrained PnP and 28 % with neuron-trained PnP (Tab. 1). W1 improves on all 60 crops under each method. Centroid error falls by 42 %, 54 % and 30 %, respectively, while PSNR rises by 1.6–1.8 dB. Improvements are present on both the 12 prespecified crops and the 48 crops sampled after parameter selection. A second optimization seed from the same starting phase gives similar gains (supplement).

Figures 1 and 5 show the effect on depth-varying neuronal features. The opening example shows reduced axial smearing along a descending dendritic feature with pretrained PnP; the all-prior views in the supplement show the corresponding TV effect. Differences are smaller with the neuron-trained prior, which already suppresses much of the diffuse signal. The views and crops were chosen from reference structure; both examples have typical-to-lower W1 gains rather than being selected for large optical improvements. Depth-slab views and a missed weak feature are shown in the supplement.

The effect is not confined to comparison with the initial phase. In the supporting encoder study on 12 crops with pretrained PnP, learned-plate centroid error is $0.230\,\mu\mathrm{m}$, compared with 0.422–$0.438\,\mu\mathrm{m}$ for three random phase plates and 0.396–$0.523\,\mu\mathrm{m}$ for three amplitude masks. Thus, the random initialization is typical of the tested phase plates. These controls assess centroid error and image quality; they do not replicate optical training from independent initial phases.

The optical and prior improvements are complementary. With the learned plate, TV attains a lower mean W1 than neuron-trained PnP with the random plate (0.218 versus $0.271\,\mu\mathrm{m}$, a paired 20 % reduction; 95 % CI: 16–23 %). This is a comparison of these two configurations, not a general ordering of optics and priors. Neuron-trained PnP with the learned plate gives the lowest mean W1 overall. Conversely, the tested in-focus DMD remains near the depth-uniform reference under all three methods (W1 2.24–$2.38\,\mu\mathrm{m}$, reference $2.27\,\mu\mathrm{m}$ on the 12 control crops), illustrating the importance of the measurement code.

6.3. Recovery of overlapping axial structures

Lower average profile error need not imply separation of overlapping structures. We therefore test controlled crossing filaments with known depth gaps (Fig. 6). The smallest gap meeting the success criterion falls from 3.84 to $2.56\,\mu\mathrm{m}$ under TV and pretrained PnP. For neuron-trained PnP, it remains $3.84\,\mu\mathrm{m}$, despite improved W1 on neuronal volumes. At a $2.56\,\mu\mathrm{m}$ gap, the learned plate allows TV to retain two axial peaks where the random plate merges their intensities more strongly.

The gain is task- and prior-dependent. Neither plate reliably resolves the three-filament control, and the zero-gap control shows no false splitting. Thus, optical learning improves the tested two-filament separation under two priors; it does not establish general neuronal layer separation. The neuronal evaluations assess depth placement and profile fidelity, while these controlled objects isolate an additional structural task.

6.4. Comparison of optical learning objectives

We compare the squared-correlation objective with a smooth worst-case coherence criterion and an amplitude-marginalized depth Fisher-information surrogate, motivated by coherence-based sensing and Fisher-based optical design [6, 21, 25]. These are implementations on the same voxel signatures and optical architecture, not reproductions of the cited microscopes. Initialization, sampling, optimizer and step budget are matched (Tab. 2).

All three criteria improve on the random plate. Across both training seeds and all three decoders, our objective gives 13–20 % lower W1 than the tested coherence and Fisher surrogates, with paired intervals excluding zero. An uncentered squared-correlation variant is essentially tied with ours under neuron-trained PnP (0.207 versus $0.206\,\mu\mathrm{m}$ W1). Thus, the evidence supports the depth-specific squared-correlation criterion under this budget; mean removal is not an established source of its advantage. Exact objective definitions and secondary results are in the supplement.

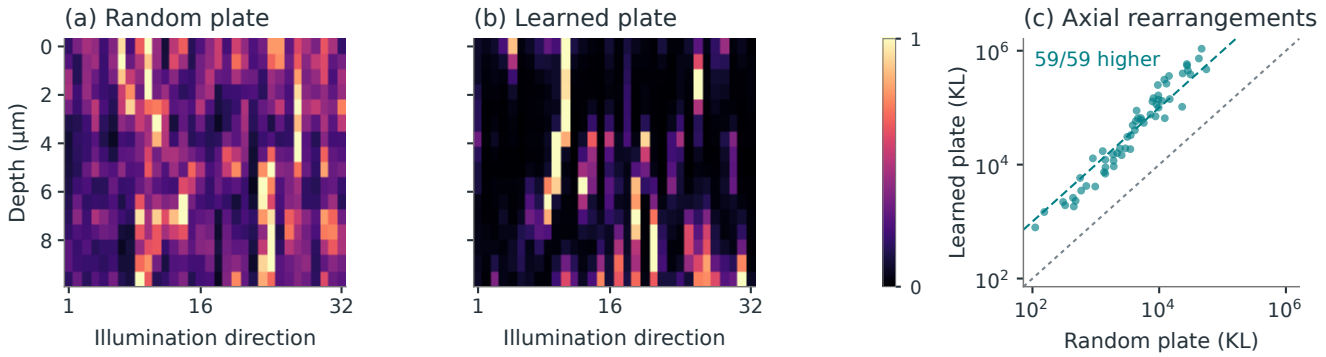


Figure 4. **Learning changes the depth code before reconstruction.** (a, b) Angular responses of the 16 depth voxels at the preset field-center position, using their corresponding camera pixel and the same 32 directions. Each row is normalized by its own maximum. The learned plate concentrates responses into depth-dependent directions. (c) Poisson measurement divergence between each neuronal volume and a merged-depth alternative preserving its projection and column centroids. Each point averages the matched direction subsets of one eligible crop; the gray diagonal indicates equal separation and the teal dashed line indicates a tenfold increase. These specified alternatives test measurement discrimination independently of a decoder.

	Axial-profile W1 ↓			Centroid error ↓			PSNR ↑	
Reconstruction	Random	Learned	Reduction [95 % CI]	Random	Learned	Reduction	Random	Learned
TV	0.371	0.218	41 % [38, 45]	0.259	0.150	42 %	36.70	38.29
Pretrained PnP	0.579	0.290	50 % [47, 53]	0.490	0.224	54 %	35.72	37.56
Neuron-trained PnP	0.271	0.195	28 % [26, 29]	0.197	0.139	30 %	38.01	39.60

Table 1. **The same learned plate improves all three reconstruction methods.** $C = 8$, matched nominal dose, 60 crops from six cells and three animals, averaging five direction subsets per crop. W1 and centroid errors are in μm; PSNR is in dB. Reductions are relative to the random initialization, with paired cell-bootstrap 95 % intervals. Neuron-trained PnP uses both slice and volumetric denoisers.

Optical objective	TV	Pretrained PnP	Neuron-trained PnP
Random plate	0.318	0.496	0.282
Smooth worst-case	0.238	0.346	0.238
Depth Fisher surrogate	0.228	0.340	0.240
Squared depth correlations	0.196	0.277	0.206

Table 2. **Matched optical training budgets.** Mean axial-profile W1 (μm) on the 12 prespecified crops at $C = 8$. All objectives use the same initial phase, sampled positions and angles, and 1000 Adam steps. Each encoder is tuned separately on validation data. The displayed training seed is 0; paired intervals, centroid errors and the second seed are in the supplement.

	Random plate		Learned plate
Reconstruction	256 images	32 images	32 images
TV	0.249	0.318	0.196
Pretrained PnP	0.413	0.496	0.277
Neuron-trained PnP	0.203	0.282	0.206

Table 3. **Fewer measurements with learned illumination.** Mean W1 (μm) on the same 12 crops at matched nominal dose. The learned plate uses 32 images ($C = 8$); the random plate uses either 256 ($C = 1$) or 32. Full compression and dose curves, centroids and paired intervals are in the supplement.

6.5. Measurement and noise budgets

The learned plate improves recovery at all three tested measurement budgets. With 32 images, it gives lower W1 than the random plate with 256 images under TV and pretrained PnP (Tab. 3). Neuron-trained PnP gives close means (0.206 versus 0.203 μm), without evidence establishing equivalence. This comparison holds nominal dose fixed by increasing exposure per image when fewer directions are acquired.

Gains also persist with fewer photons. At 0.3 times the nominal dose, the learned plate gives 22–39 % lower W1 than the random plate at full dose across the three methods. When both encoders use one tenth of the dose, the reductions are 31–51 %. Adding uniform fluorescence background and Gaussian read noise retains 27–53 % reductions, with scalar decoder parameters retuned on validation data for each condition. These results concern the simulated dose and noise model; no acquisition-speed or phototoxicity measurement is inferred.

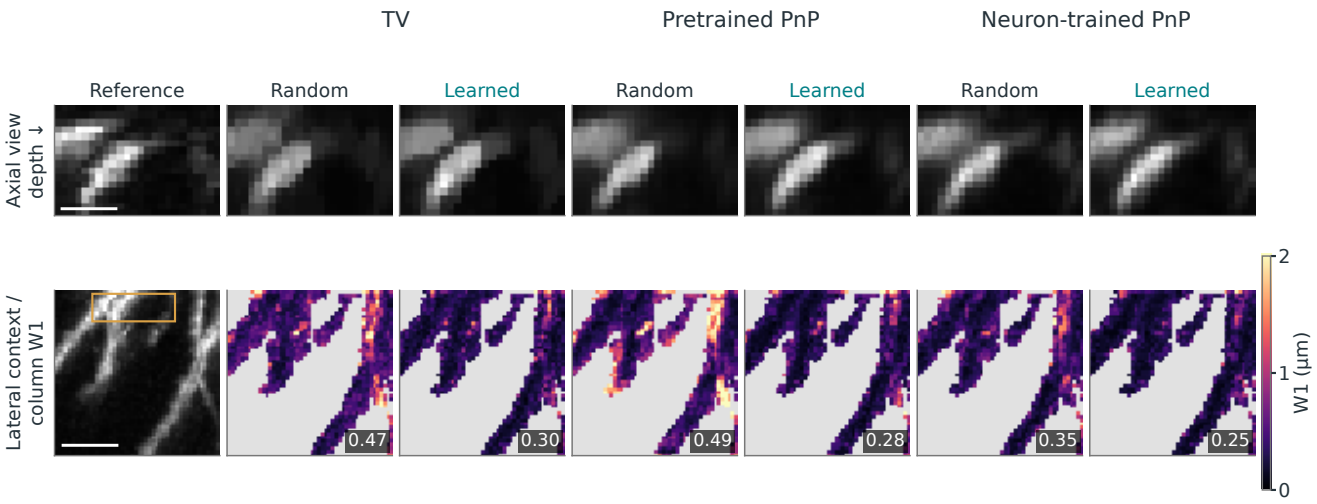


Figure 5. **Where axial errors change in a second neuronal volume.** Top: matched axial band views for all three priors. Bottom: the reference lateral projection and column-wise W1 maps; gray marks columns excluded by the reference signal mask. This companion was selected from another animal using the same reference-only rule. One display scale is shared across each row. Panel values are draw-0 signal-weighted crop W1 (μm). Bars: 5 μm (axial view), 10 μm (lateral context).

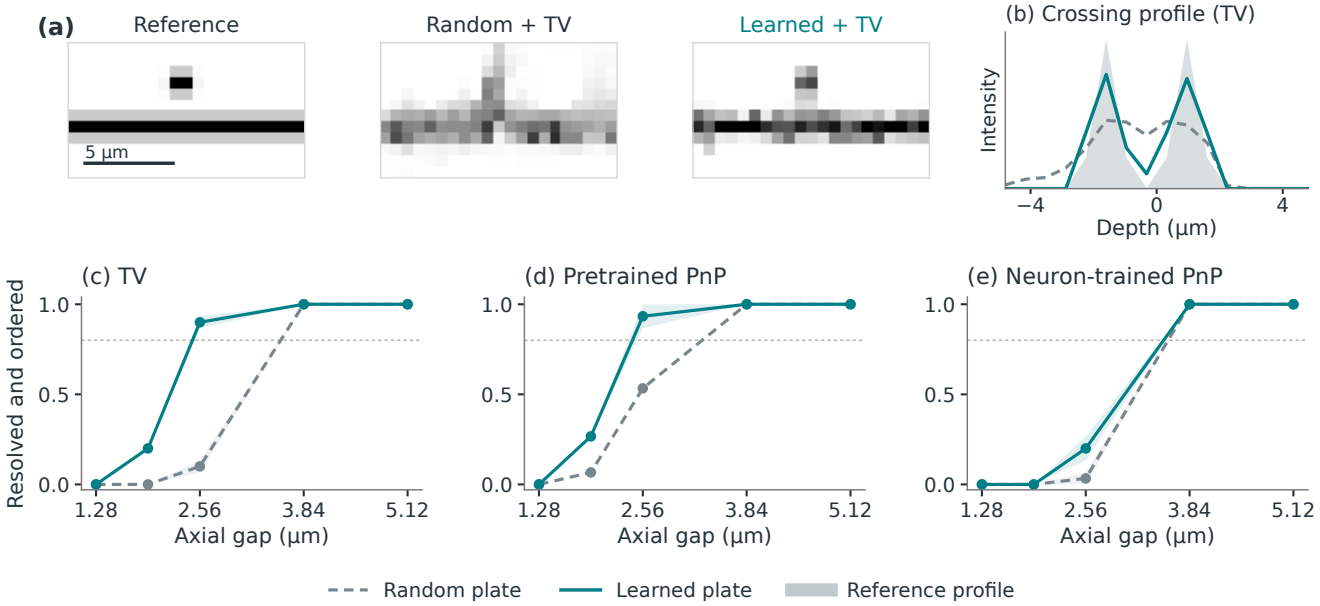


Figure 6. **Controlled axial separation is prior-dependent.** (a) Matched sections through a crossing, using one intensity scale. (b) Mean profile over its four crossing columns. The displayed gap is the learned plate's TV threshold; geometry, center and draw are the first fixed case. (c–e) Success rates across three geometries and five draws at each of two axial centers. Lines pool centers; bands span their rates, rather than confidence intervals. The 80 % threshold must hold at each center separately. Bar: 5 μm.

6.6. Alignment tolerance and calibration

Accurate optical registration is important for the learned code. Under neuron-trained PnP at $C = 8$, plate-distance offsets of $\pm 0.5\,\mu$m retain 27–28 % W1 reductions when the operator is rebuilt using the known offset and scalar reconstruction parameters are revalidated. The plate itself is not retrained. Pointing errors of 0.005–0.01 in direction cosine retain 13–20 % reductions with the nominal operator.

Unmodeled registration errors can reverse the benefit. With a nominal operator, the same distance offsets make learned-plate W1 4–6 % worse than the random plate; a lateral shift of one plate feature makes it 11 % worse (95 %

CI: 2–23 %). The simulations therefore support tolerance to characterized distance changes, not robustness to arbitrary misalignment. Estimating the realized operator is an essential step for a physical implementation.

7. Discussion and limitations

Learning the measurements. The same optical design improves axial recovery under TV and two PnP priors, although no decoder or specimen enters plate training. This supports designing depth discrimination as a property of the measurements, then choosing the reconstruction method separately. Prior choice still matters: neuron-trained PnP gives the lowest mean W1 with the learned plate, and the two PnP variants share an ADMM framework. The results establish benefit across the tested methods, not arbitrary decoders or superiority over end-to-end learning, which we do not compare.

What improves in the recovered volume. Full-profile W1 and centroid error both decrease, indicating improved axial placement without relying on centroid accuracy alone. Their relative gains are similar; the evidence does not show that profile structure improves more than its centroid. Controlled crossings provide a more specific separation result under TV and pretrained PnP, with no threshold improvement under neuron-trained PnP. These findings support better depth recovery and some controlled structural separation, rather than general neuronal layer recovery or identifiable connectivity. Supplementary measurement-consistency bounds remain uninformative on the tested dense neuronal windows.

Physical implementation. All results are simulated under linear fluorescence in a homogeneous medium, using idealized optical components and a common excitation/emission wavelength. Finer-grid measurement generation preserves specimen discretization error, but uses the same optical physics as reconstruction. Scattering, aberrations, wavelength differences and unknown system calibration remain untested. The alignment results make operator characterization a practical requirement: known distance changes can be accommodated, while unmodeled registration errors can reverse the optical gain. A physical experiment must characterize the realized patterns and detection response before evaluating reconstruction.

Extent of the evidence. The 60 crops come from six cells in three animals, so the cell-bootstrap intervals remain approximate and do not represent broad biological replication. Optical training uses two sampling seeds from one initial phase; independent initializations have not been tested. The additional random-plate and amplitude-mask controls support the choice of baseline but do not close that gap. Objective comparisons use a matched step budget for specific surrogates, without an exhaustive search over alternative design criteria. These boundaries define the scope of the demonstrated optical improvement.

8. Conclusion

We learn a fixed phase plate for angle-scanned, fixed-focus fluorescence imaging by optimizing detector-integrated depth signatures, without specimen examples or a reconstruction solver. In simulations, the learned plate reduces axial-profile error by 28–50 % and centroid error by 30–54 % across TV and two PnP methods. Controlled crossings show improved separation under two of the three priors. These results demonstrate that optical learning can improve axial recovery across reconstruction methods while leaving the reconstruction prior outside the training loop.

LLM Usage Statement

OpenAI ChatGPT/Codex assisted with manuscript organization, drafting and editing, literature checks, and figure/table code. Numerical panels use simulation outputs. The authors retain responsibility for the methods, results, citations and final manuscript.

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